

RUNTIME DIVERGENCE ANALYSIS

A Methodology for Assessing Representational Correspondence
in Complex Institutional System

Methodology Specification • Version № 1.0 • First Publication

Mohamed Sultan Mohamed

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ABSTRACT

This document presents the formal specification of Runtime Divergence Analysis (RDA), a methodology for establishing the functional architectural correspondence between the interface-layer signal emissions of a complex institutional system and the runtime execution behavior actually produced by the system architecture. RDA establishes that every system making representational claims about its operations generates two empirically distinguishable layers of observable outputs: A.) interface signals, and B.) execution artifacts. The correspondence between these layers is a measurable structural property verifiable independent to outside, or external, system disclosure. This specification articulates the foundational principles, definitional framework, and eight-phase methodological procedure through which RDA produces self-authenticating, reproducible governance findings applicable across any institutional domain in which representational claims are made about operational functionality.

1 Statement of the Problem

Modern institutional systems, including digital platforms, credentialing programs, financial products, healthcare applications, and AI systems, routinely make representational claims about their functional architecture: claims about what they do, how they do it, and what participants can expect when they engage. These claims are communicated through interface-layer signals: marketing language, user interface design, platform documentation, product descriptions, and statements of authorized representatives.

The central problem this methodology addresses is the absence of a systematic, reproducible procedure for establishing whether interface-layer signals correspond to the runtime execution behavior of the system they describe. Without such a procedure, representational claims cannot be evaluated against operational reality except through internal institutional disclosure, subject to institutional interests, or through regulatory investigation with discovery powers. Neither mechanism is available to the ordinary participant relying on the representation at the point of enrollment, purchase, or engagement.

The problem presents in three structural dimensions:

The Evidentiary Dimension: Claims about operational architecture require primary-source evidence typically held by the institution, creating an asymmetric information condition in which the party making the claim controls the evidence needed to test it.

The Structural Dimension: Many representational divergences are architectural, they describe systems that were never built, not systems that subsequently failed, making individual outcome evidence insufficient to establish the structural nature of the gap.

The Accountability Dimension: When representational divergence is structural, responsibility cannot be assigned to individual actors because the divergence is a feature of system design rather than of any individual decision. Existing accountability frameworks are calibrated for individual conduct, not architectural design.

RDA resolves all three dimensions simultaneously. It generates primary-source evidence using the institution's own authenticated artifacts, identifies divergences at the structural rather than the outcome level, and produces findings independent of individual intent and attributable to system architecture rather than to individual actors.

4 The Runtime Divergence Analysis Methodology

The RDA methodology consists of eight sequential phases. Each phase produces a documented artifact record that contributes to the cumulative evidentiary architecture. Phases are executed in sequence where findings from earlier phases provide the evidentiary baseline for subsequent phases. The methodology is complete when all eight phases have been executed and documented.

Phase 1 — Interface-Layer Signal Inventory

Objective: Establish a complete, sourced catalog of all interface-layer signals relevant to the system under examination.

Procedure:

1. Identify all channels through which the system communicates representations of its functional architecture: marketing materials, product documentation, UI/UX elements, and representative statements.
2. For each identified channel, catalog the specific signals and their source attribution, date, channel, and authorized representative where applicable.
3. Classify each signal by functional characteristic implied: mechanism claims (the system does X), pathway claims (the system provides access to Y), integration claims (the system connects X to Y), or outcome claims (the system produces Z).
4. Record the evidentiary source for each signal with sufficient specificity to allow independent verification by any observer.

Output: A sourced Interface-Layer Signal Inventory classifying each signal by type and documenting its provenance and evidentiary source.

Phase 2 — Runtime Artifact Collection Protocol Design

Objective: Design the specific engagement sequence that will generate runtime execution artifacts relevant to each ILS identified in Phase 1.

Procedure:

1. For each ILS implying functional characteristics, identify the engagement sequence a reasonable actor relying on that signal would follow.
2. Design the artifact collection protocol to capture observable state at each step of the engagement sequence, with particular attention to system boundary crossing events.
3. Identify which system infrastructure will authenticate each artifact in the collection sequence.
4. Document the protocol in sufficient detail to permit independent reproduction by any observer against the same system.

Output: A documented Runtime Artifact Collection Protocol specifying the engagement sequence and artifact documentation procedure for each relevant ILS.

10 Prior Art and Publication Notice

Publication and Prior Art Statement

This methodology specification establishes public prior art for the Runtime Divergence Analysis (RDA) methodology. The foundational principles, definitional framework, and methodological procedure described herein constitute the original intellectual contribution of the author, developed through primary-source investigation conducted during the period from February through April 2026. The worked example referenced in Section 9 constitutes the first documented field application of the methodology.

Any subsequent application, adaptation, extension, or implementation of this methodology in any domain should reference this specification as the foundational document. The author retains intellectual priority in the foundational principles and methodological procedure as stated herein, independent of any institutional engagement or consulting arrangement entered into subsequent to this publication.

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